

Megan E. Hillis, Ph.D.

meg.hillis12@gmail.com | megan-hillis.github.io

ACADEMIC APPOINTMENTS

Postdoctoral Researcher & Lecturer Dartmouth College, Cognitive Science Department	2025-
Graduate Student in Cognitive Neuroscience Dartmouth College, Psychological and Brain Sciences Department	2019-2025
Research Assistant State University of New York at Geneseo, Department of Psychology	2016-2019

EDUCATION

Ph.D. in Cognitive Neuroscience Dartmouth College, Guarini School of Graduate and Advanced Studies Advisor: Dr. David J. M. Kraemer Dissertation: <i>Neural patterns reflect semantic representation in novice sign language learners</i>	2019-2025
M.S. in Psychological and Brain Sciences Dartmouth College, Guarini School of Graduate and Advanced Studies Advisor: Dr. David J. M. Kraemer Thesis: <i>Overlapping semantic representations of sign and speech in novice sign language learners</i>	2024
B.A. in Psychology with Honors, Minor in German Language State University of New York at Geneseo GPA: 3.94, <i>summa cum laude</i> Thesis Advisor: Dr. Jeffrey Mounts Honors Thesis: <i>The Dorsal Visual Stream and Object Recognition: Weighing the evidence for dorsal category sensitivity</i>	2015-2019

PUBLICATIONS

Preprints:

Hillis, M. E., & Kraemer, D. J. M. (under review). Initial signs of learning: Decoding newly-learned vocabulary from neural patterns in novice sign language learners. *bioRxiv*.
<https://doi.org/10.1101/2025.04.11.648265>

Published:

Hillis, M.E. & Kraemer, D.J.M. (2026) From seeing to understanding: Novice sign language learners shift focus from perceptual to semantic information for newly learned signs. *Proceedings of the Annual Meeting of the Cognitive Science Society*. 48 (48).

Hillis, M.E.*, Cetron, J.S.*, Diamond, S.G., May, V.V.; & Kraemer, D. J. M.. (2026) Neural patterns reflect conceptual grasp of novice students following first class learning in physics. *npj Sci. Learn.* 11, 20. <https://doi.org/10.1038/s41539-025-00394-3>

*co-authors contributed equally to this work and are listed alphabetically by the journal

Blanchet, J. B., **Hillis, M. E.**, Lee, Y., Shao, Q., Zhou, X., Balkcom, D., & Kraemer, D. J. M. (2025). Enhancing the Educational Potential of Online Movement Videos: System Development and Empirical Studies with TikTok Dance Challenges. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–19. <https://doi.org/10.1145/3706598.3714062>

Blanchet, J., **Hillis, M. E.**, Lee, Y., Shao, Q., Zhou, X., Kraemer, D. J. M., & Balkcom, D. (2023). LearnThatDance: Augmenting TikTok Dance Challenge Videos with an Interactive Practice Support System Powered by Automatically Generated Lesson Plans. *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 1–4. <https://doi.org/10.1145/3586182.3615801>

Blanchet, J., **Hillis, M.E.**, Lee, Y., Shao, Q., Zhou, X., Kraemer, D.J.M., Balkcom, D. (2023) Automatic Generation and Teaching of Dance Lessons from Video. *Proceedings of the 24th International Workshop on Mobile Computing Systems* (145-145)

Hillis, M.E., Aubrey, B., Blanchet, J., Shao, Q., Balkcom, D., Zhou, X., & Kraemer, D. J. M. (2022) Overlapping semantic representations of sign and speech in novice sign language learners. *Proceedings of the Annual Meeting of the Cognitive Science Society*. 44 (44)

Alfred, K. A., **Hillis, M. E.**, & Kraemer, D. J. M. (2020) Individual differences in the neural localization of relational networks of semantic concepts. *Journal of Cognitive Neuroscience*, 3(33), 1-12

Shao, Q., Sniffen, A., Blanchet, J., **Hillis, M.**, Shi, X., Haris, T., Liu, J., Lamberton, J., Malzkahn, M., Quandt, L., Mahoney, J., Kraemer, D., Zhou, X., & Balkcom, D. (2020). Teaching American Sign Language In Mixed Reality. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*. 4(4), 1-27

TALKS

Learning and Brain Development Lab Trainee Seminar, Boston, MA July 2026
From seeing to understanding: Novice sign language learners shift focus from semantic to perceptual information

Cognitive Science Weekly Speaker Series, Hanover, NH May 2025
Decoding newly learned vocabulary from neural patterns in novice sign language learners

Cognitive Science Weekly Speaker Series, Hanover, NH Oct 2022
Characterizing changes in knowledge and learning with multivariate fMRI

Cognitive Neuroscience Society Data Blitz, San Francisco, CA

Mar 2022

Decoding knowledge of newly-learned language from neural representations of semantic meaning

GREAT Day Symposium, Geneseo, NY

May 2019

The Dorsal Visual Stream and Object Recognition: Weighing the evidence for dorsal category sensitivity

POSTERS

Hillis, M.E., & Kraemer, D.J.M. (2026) From seeing to understanding: Novice sign language learners shift focus from perceptual to semantic information for newly learned signs. *Cognitive Science Society*, Rio de Janeiro, Brazil (Hybrid Presentation)

Hillis, M.E., & Kraemer, D.J.M. (2026) Reading the Signs: Encoding models generalize from English to ASL in novice sign learners. *Cognitive Neuroscience Society*, Vancouver, British Columbia, CAN.

Hillis, M.E. & Kraemer, D.J.M. (2025) Neural patterns reflect quiz performance in novice sign language learners. *Cognitive Neuroscience Society*, Boston, MA, USA.

Hillis, M.E., Shao, Q., Lee, Y., Blanchet, J., Zhou, X., Balkcom, D., & Kraemer, D.J.M. (2024) Using biosignals to predict cognitive load in an adaptive learning task. *International Mind, Brain, and Education Society*, Leuven, Belgium.

Hillis, M.E., Aubrey, B., Blanchet, J., Shao, Q., Balkcom, D., Zhou, X., & Kraemer, D. J. M. (2024). Neural decoding of semantic representations from novice sign language learners reflects newly-acquired vocabulary. *Cognitive Neuroscience Society*, Toronto, Ontario CAN.

Hillis, M.E., Aubrey, B., Blanchet, J., Shao, Q., Balkcom, D., Zhou, X., & Kraemer, D. J. M. (2023). Decoding knowledge of newly-learned language from neural representations of semantic meaning. *Cognitive Neuroscience Society*, San Francisco, CA, USA.

Hillis, M.E., Aubrey, B., Blanchet, J., Shao, Q., Balkcom, D., Zhou, X., & Kraemer, D. J. M. (2022) Overlapping semantic representations of sign and speech in novice sign language learners. *Cognitive Science Society*, Toronto, Ontario CAN

Hillis, M.E., Aubrey, B., Blanchet, J., Shao, Q., Balkcom, D., Zhou, X., & Kraemer, D. J. M. (2022). Sign and speech share semantic representation in novice sign language learners. *International Mind, Brain, and Education Society*, Montreal, Quebec CAN

Alfred, K., **Hillis, M.E., & Kraemer, D.** (2020) Individual Differences in Patterns of Semantic Distance. *Society for the Neuroscience of Creativity*, Boston MA, USA

Mounts, J., Edwards, A., Matyasovszky, G., **Hillis, M.E.** & Nguyen, A. (2017) *The Effects of Endogenous Attention on Spatial and Temporal Processing*. GREAT Day Symposium, Geneseo NY, USA

TEACHING

Dartmouth College – Sole Instructor

- Modeling Mind and Behavior (COGS 50.08) Fall 2025

Dartmouth College Teaching Apprenticeships and Lectures

- Major Seminar in Cognitive Science (Guest Lecturer) Fall 2022, Spring 2025
- Principles of Human Brain Mapping with fMRI (TA, Lab Instructor) Spring 2022
- Introduction to Neuroscience (TA, Guest Lecturer) Fall 2021
- Laboratory in Psychological Science (TA, Guest Lecturer, Lab Instructor) Summer 2020 & 2021

PROFESSIONAL ACTIVITIES AND SERVICE

Workshops and Trainings

- Methods In Neuroscience at Dartmouth (MIND) Summer School Aug 2023
- DCAL Future Faculty Teaching Series (8-week workshop) Jan-Mar 2021

Dartmouth College Service

- Graduate Research Roundtable (Co-organizer) 2021-22
- Psychological and Brain Sciences Social Committee (Volunteer) Ad Hoc, 2021-25
- PBS Dept. Graduate Peer Mentor 2021-23
- PBS Dept. Working Group Advisory Committee (Member) 2021

STEM Education Outreach

- Dartmouth Science Day (Volunteer) 2026
- Letters to a Pre-Scientist (Volunteer) 2025
- Nature Education Specialist: Girl Scouts of Northeastern New York 2019
(designed & led STEM programs for ages 5-18)

AWARDS & SOCIETIES

Edward Curry Scholarly Achievement Award	Awarded 2019
Edgar Fellows – Geneseo Merit Scholars	Inducted 2016
Phi Eta Sigma – First Year Honor Society	Inducted 2016
Phi Beta Kappa – Honor Society	Inducted 2019
Psi Chi – Psychology Honor Society	Inducted 2019

SKILLS

Programming & Analysis:	Python (8 years' experience) R (3 years' experience) fMRI analysis tools including AFNI, FSL, fMRIPrep, FreeSurfer
Languages:	English (native) German (CEFR B2 Level)
Other:	OpenBCI mobile biosensing devices Experimental design and implementation with PsychoPy, Qualtrics, Google Office Tools (including Google Apps scripting), Amazon Mechanical Turk Adobe Photoshop